

STATISTICAL FUNCTIONS

AIM:

To compute basic statistical measures (mean, median, mode, variance, and standard deviation), and visualize different data distributions using histograms.

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import random

df=[random.randint(1,100) for x in range(100)]
df

df=pd.DataFrame(df,columns=['Values'])
df

mean_arr=df['Values'].mean()
median_arr=df['Values'].median()
mode_arr=df['Values'].mode()[0]
var_arr=df['Values'].var()
std_arr=df['Values'].std()

print(f'mean: {mean_arr}')
print(f'median: {median_arr}')
print(f'mode: {mode_arr}')
print(f'variance: {var_arr}')
print(f'standard deviation: {std_arr}')

plt.hist(df['Values'],bins=10,edgecolor='black')
plt.xlabel('Values')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()

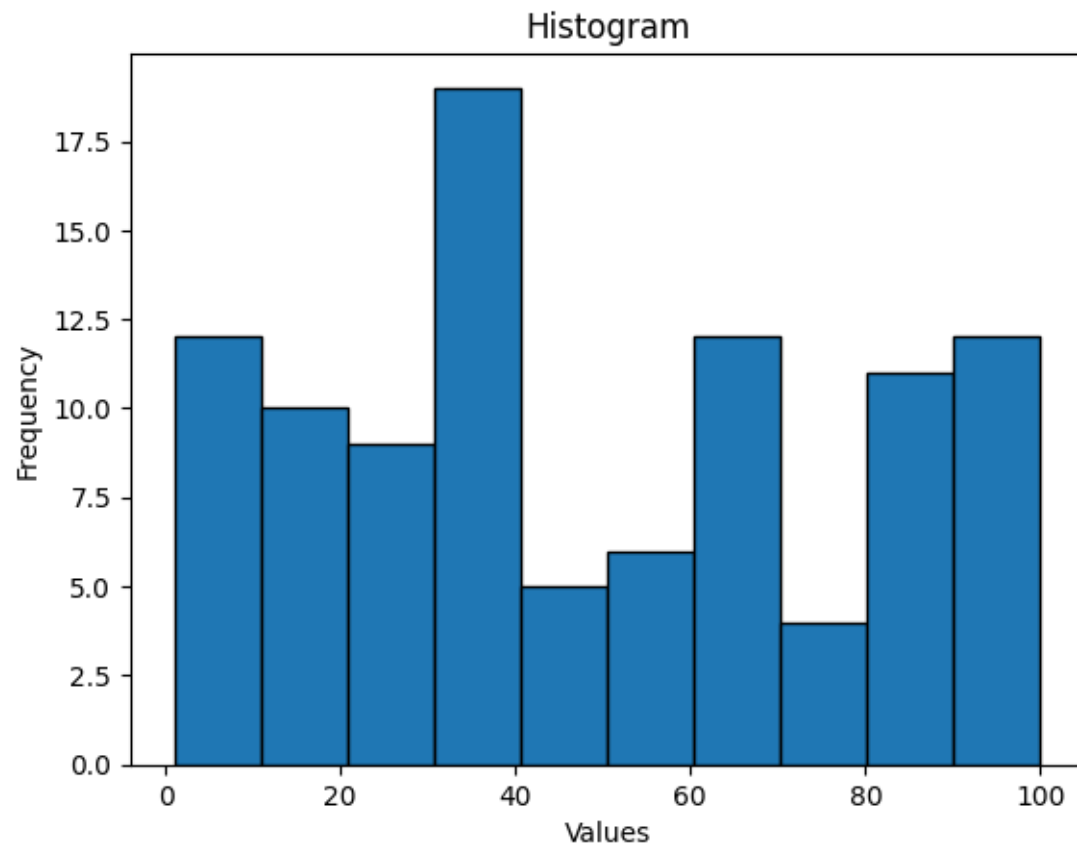
sample_uni=np.random.uniform(low=0,high=100,size=100)
plt.hist(sample_uni,bins=10,edgecolor='black',alpha=0.5,label='Uniform')
plt.xlabel('Values')
plt.ylabel('Frequency')
plt.title('Uniform Distribution')
plt.legend()
plt.show()

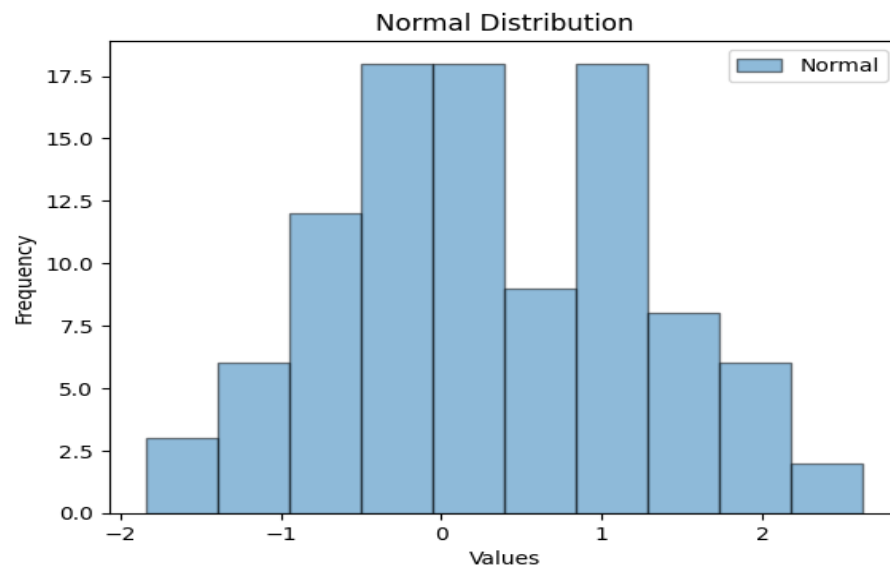
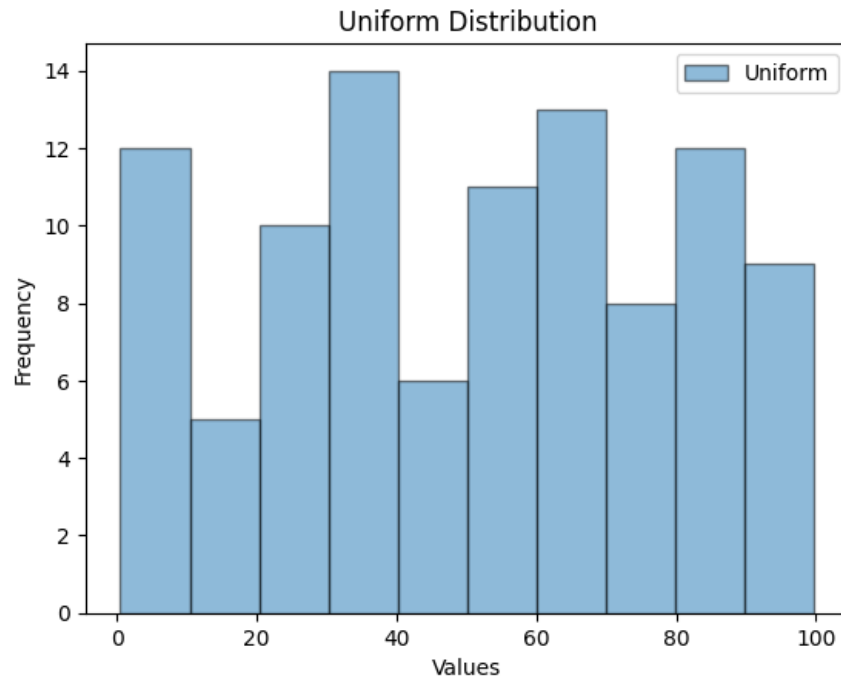
sample_norm=np.random.normal(loc=0,scale=1,size=100)
plt.hist(sample_norm,bins=10,edgecolor='black',alpha=0.5,label='Normal')
plt.xlabel('Values')
plt.ylabel('Frequency')
```

```
plt.title('Normal Distribution')
plt.legend()
plt.show()
```

```
df['Normal']=sample_norm
df['Uniform']=sample_uni
df
```

OUTPUT:





RESULT:

The program successfully generated random data and calculated key statistical measures such as mean, median, mode, variance, and standard deviation. Histograms were plotted to visualize the distribution of the dataset, as well as uniform and normal distributions.